

BOEKBESPREKINGS / BOOK REVIEWS

Introduction to Biostatistics

by Robert R. Sokal and F. James Rohlf

First Edition, xiii + 368 pages, 52 figures and 14 tables. W.H. Freeman and Company, San Francisco, California, 1972. Price: \$9,50

The authors set themselves the task of writing an introductory text on biostatistics for the undergraduate student requiring no more than an elementary preparation in mathematics. The book is also intended for short courses in biostatistics in medical and other professional schools.

To a large extent the authors have been successful.

Chapter 1 is a short introduction giving definitions and summarising the development of biostatistics. Chapter 2 covers data collection for both discrete and continuous cases. Special tables called "Boxes" are introduced. These illustrate computational methods for solving various types of biostatistical problems and may therefore be used as convenient "patterns". This feature is used throughout the book. Chapter 3 contains a treatment of descriptive statistics including the mean, the geometric mean, median, mode and the range. Chapter 4 contains the Binomial and Poisson distributions while the normal distribution is treated in Chapter 5. Chapter 5 deals with the testing of different hypotheses. Students' *t* and the chi-square distribution are also introduced. In Chapters 7-9 the concept of the analyses of variance is presented in a very satisfactory manner. In Chapter 10 two non-parametric methods are discussed in lieu of ANOVA. Chapters 11 and 12 contain a good treatment of regression and correlation. Tests of significance in regression and correlation are discussed. Chapter 13 deals with the analysis of frequencies. Tests for goodness of fit are dealt with. A *G*-statistic ($G = 2 \ln L$ where *L* is the likelihood function) is employed in lieu of the traditional chi-square test.

The emphasis in this book is on the practical applications of statistics to biology, as such it forms an excellent, easily readable, and compact text, well worth a place on one's bookshelf.

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Understanding Statistical Reasoning

by Eleanor W. Willemsen

First Edition, xi + 223 pages, 28 illus., 67 tables, W.H. Freeman and Company, San Francisco, 1974. Price: \$4.95 (U.K. £5.20 cloth, £2.40 paper)

The subtitle of this book is "How to evaluate research literature in the behavioural sciences" and, as is stated in the preface, has been written for those whose primary need

for understanding statistical reasoning arises in their critical reading and interpretation of the research literature of the behavioural sciences. Consequently, this is not a text book on statistical theory, but takes over where most textbooks and statisticians (!) leave off – that is at the interpretation of the statistical calculations.

Using the minimum of mathematics, Dr Willemsen discusses most of the common statistical techniques – descriptive statistics, hypothesis testing, correlation, analysis of variance, multivariate analysis and non-parametric statistics. By using simple examples from the behavioural sciences, the author shows how the results of these statistical techniques can give an insight into the natural phenomena under investigation, but perhaps more importantly, she also shows how the same techniques can sometimes obscure the truth that lies beneath. This last aspect is particularly well illustrated in the discussion of Pearson's correlation coefficient, where the author gives reasons for unexpectedly high or low correlations. Realising that when non-parametric procedures are used to evaluate experimental results, the research report often contains only scant reference to the method used, Dr Willemsen provides a useful glossary of the most commonly encountered non-parametric statistical tests with examples of their use.

In the reviewer's opinion, this book admirably fills a gap in statistical literature, and could be recommended as additional reading for almost any statistics course. There are a few obvious misprints in formulae given on pages 61 and 72.

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